

Elektroniske enheter og kretser

Lab 03: BJT Common-Emitter and Class-A Power Amplifiers

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Abstract—This report presents an investigation of BJT common-emitter and class-A power amplifiers. The work is divided into two parts. In the first part, the DC and AC behaviour of a common-emitter amplifier is studied, including bias voltages, voltage gain, input impedance, and output impedance, with measured results compared against calculated values. In the second part, the circuit is reconfigured as a class-A power amplifier, and its input power, output power, and efficiency are evaluated under maximum undistorted output conditions. The measurements were generally in good agreement with theoretical calculations, within about ten per cent for bias and impedance parameters. For the power and efficiency analysis, larger discrepancies were observed, mainly due to transistor non-idealities and practical measurement limitations. The study demonstrates how theoretical analysis and experimental results can be combined to assess the performance and limitations of amplifier circuits.

Index Terms—BJT amplifier, common-emitter, class-A, voltage gain, impedance, efficiency

I. INTRODUCTION

The BJT common-emitter amplifier is a common transistor circuit, known for providing voltage gain with phase inversion and moderate input and output impedances. A clear understanding of its DC bias point and AC characteristics is important for predicting circuit behaviour and for practical design. The class-A power amplifier is a related circuit that operates with the transistor biased in its linear region, enabling it to deliver an undistorted output at the cost of efficiency. This report investigates both configurations as part of a laboratory assignment on BJT common-emitter and class-A power amplifiers. The objectives are to measure AC and DC voltages in a common-emitter amplifier, to determine voltage gain, input impedance, and output impedance, and to calculate and measure input and output power for a class-A amplifier. The analysis combines theoretical calculations with experimental measurements in order to compare expected and observed performance, and to highlight the effects of practical limitations.

II. PART 1-4

A. Common Emitter DC Bias

The first step of the analysis was to measure the resistor values used in the circuit. The results are summarised in Table I.

TABLE I
MEASURED RESISTOR VALUES

Parameter	Value	Unit
R_E	0.9981	k Ω
R_C	3.0008	k Ω
R_2	9.9911	k Ω
R_1	33.423	k Ω

Using these resistance values, the approximate operating point of the transistor can be calculated. Since the base current I_B is much smaller than the current through R_2 , the base voltage is approximated as shown in Eq. (1).

$$V_B \approx V_{CC} \cdot \frac{R_2}{R_1 + R_2} \quad (1)$$

The emitter voltage is then given by Eq. (2), assuming a base-emitter drop of 0.7 V.

$$V_E = V_B - V_{BE} \quad (2)$$

The emitter current is determined from Ohm's law according to Eq. (3).

$$I_E = \frac{V_E}{R_E} \quad (3)$$

As $I_C \gg I_B$, the collector current is approximated as equal to the emitter current. The collector voltage is therefore given by Eq. (4).

$$V_C = V_{CC} - (I_C \cdot R_C) \quad (4)$$

Finally, the intrinsic emitter resistance r_e is calculated using Eq. (5), which is provided in the assignment as Eq. 3.1.

$$r_e = \frac{25 \text{ mV}}{I_E} \quad (5)$$

The results of these calculations are listed in Table II.

TABLE II
CALCULATED NODE VOLTAGES AND EMITTER CURRENT

Parameter	Value	Unit
V_{CC}	10.000	V
V_B	2.287	V
V_E	1.587	V
V_C	5.228	V
I_E	1.590	mA
r_e	15.722	Ω

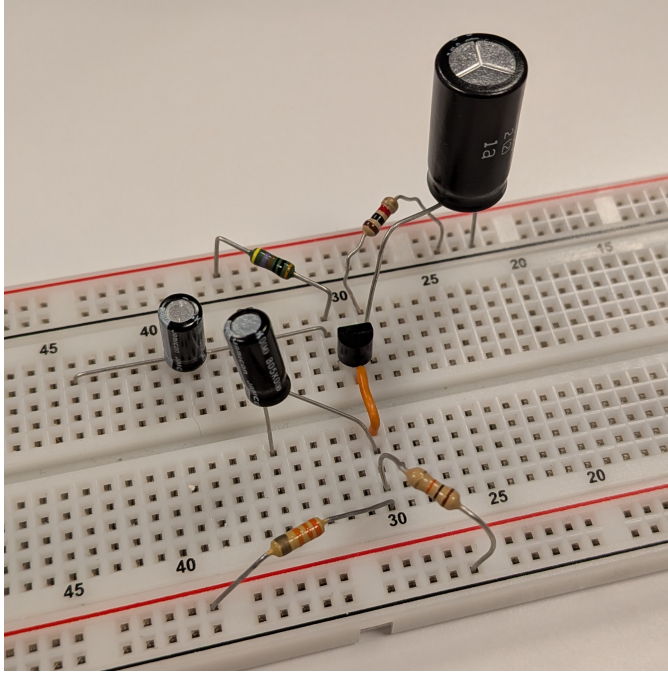


Fig. 1. Photograph of the constructed Voltage-Divider Bias amplifier circuit.

The constructed circuit is shown in Figure 1. The measured node voltages were recorded, and the corresponding values of I_E and r_e were calculated, as shown in Table III.

TABLE III
MEASURED NODE VOLTAGES AND EMITTER CURRENT

Parameter	Value	Unit
V_{CC}	9.98	V
V_B	2.215	V
V_E	1.541	V
I_E	1.544	mA
V_C	5.38	V
r_e	16.192	Ω

To compare measured and theoretical results, the relative difference is calculated using Eq. (6). In this expression, M denotes the measured value and C denotes the calculated (theoretical) value. The difference is expressed as a percentage of the calculated value.

$$\text{Difference (\%)} = \frac{M - C}{C} \cdot 100 \quad (6)$$

For this part of the experiment, only the intrinsic emitter resistance r_e was compared. The measured value of r_e differed from the calculated value by 2.99%.

B. Voltage Gain

The voltage gain of the amplifier can be estimated theoretically using the intrinsic emitter resistance. According to the assignment, the expression for A_V is given in Eq. (7), which is referred to as Eq. 3.2.

$$A_V = -\frac{R_C}{r_e} \quad (7)$$

The input signal specified in the assignment is $V_{\text{sig}} = 20 \text{ mV}$. However, this value resulted in a distorted output, so the signal generator was set to 14 mV to avoid distortion.

The output amplitude V_o was measured and the experimental voltage gain was calculated using Eq. (8). This expression follows directly from the definition of voltage gain as the ratio of output to input. The negative sign indicates the 180° phase inversion of the signal, which also allows direct comparison with the theoretical gain.

$$A_V = -\frac{V_o}{V_{\text{sig}}} \quad (8)$$

The calculated and measured values are summarised in Table IV. The relative difference in A_V was calculated using Eq. (6).

TABLE IV
CALCULATED AND MEASURED VOLTAGE GAIN

Parameter	Value	Unit
V_{sig}	13.7	mV
V_o	2.14	V
A_V (calc.)	-185.321	—
A_V (meas.)	-156.204	—
Difference	-15.71	%

C. Input Impedance

The input impedance of the amplifier can be estimated theoretically using Eq. (9), which is referred to as Eq. 3.3 in the assignment. The data sheet specifies a transistor current gain in the range 150–400. An approximate value of $\beta = 250$ was chosen for this calculation.

$$Z_i = R_1 \parallel R_2 \parallel (\beta \cdot r_e) \quad (9)$$

The input impedance was also determined experimentally by inserting a series resistor of 1 k Ω at the input and setting the signal generator to 20 mV. From the resulting division of voltage between Z_i and R_x , the input impedance is obtained according to Eq. (10).

$$Z_i = \frac{V_i}{V_{\text{sig}} - V_i} \cdot R_x \quad (10)$$

The results of the theoretical and experimental calculations are summarised in Table V. The relative difference in Z_i was calculated using Eq. (6).

TABLE V
CALCULATED AND MEASURED INPUT IMPEDANCE

Parameter	Value	Unit
R_x	0.9863	$\text{k}\Omega$
V_{sig}	19.7	mV
V_i	14.0	mV
Z_i (calc.)	2.6466	$\text{k}\Omega$
Z_i (meas.)	2.4225	$\text{k}\Omega$
Difference	-8.47	%

D. Output Impedance

The output impedance is first estimated theoretically using Eq. (11), which corresponds to Eq. 3.4 in the assignment.

$$Z_o = R_C \quad (11)$$

The experimental value of Z_o was obtained by loading the output with a resistor and using the voltage-divider relationship between Z_o and R_L . Following the same approach as for Z_i , the expression in Eq. (12) is derived from the divider that produces the loaded output V_L from the unloaded V_o .

$$Z_o = \frac{V_o - V_L}{V_L} \cdot R_L \quad (12)$$

The signal generator was set to 14 mV to avoid distortion, with a load of $R_L = 3 \text{ k}\Omega$ connected. The measured quantities and the resulting impedances are summarised in Table VI. The relative difference in Z_o was calculated using Eq. (6).

TABLE VI
CALCULATED AND MEASURED OUTPUT IMPEDANCE

Parameter	Value	Unit
R_L	3.001	k Ω
V_{sig}	13.7	mV
V_o	2.14	V
V_L	1.08	V
Z_o (calc.)	3.0008	k Ω
Z_o (meas.)	2.9454	k Ω
Difference	-1.85	%

III. POWER AND EFFICIENCY ANALYSIS

A. DC Bias Analysis

The circuit is rewired to use a different transistor and resistor values. The measured resistor values are listed in Table VII.

TABLE VII
MEASURED RESISTOR VALUES FOR PARTS 5-6

Parameter	Value	Unit
R_E	20.07	Ω
R_C	134.02	Ω
R_2	176.2	Ω
R_1	986.6	Ω

The bias point was estimated using the same relations introduced earlier, namely Eqs. (1)–(4). The resulting calculated values are summarised in Table VIII.

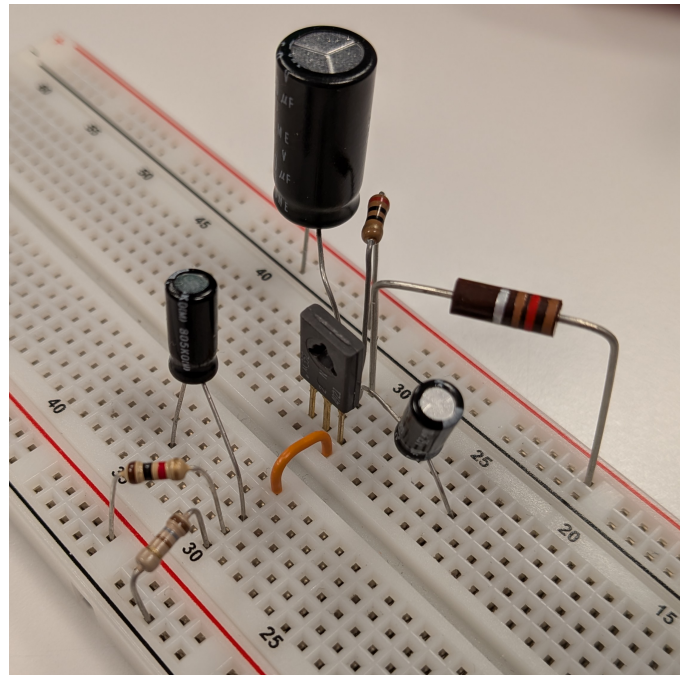


Fig. 2. Photograph of the constructed Class-A amplifier circuit.

TABLE VIII
CALCULATED DC BIAS VALUES

Parameter	Value	Unit
V_{CC}	10.000	V
V_B	1.515	V
V_E	0.815	V
V_C	4.556	V
I_E	40.623	mA

The circuit was then constructed using the measured resistors. A photograph of the assembled circuit is shown in Figure 2. From this circuit, the node voltages were measured and the emitter current was derived. The results are presented in Table IX.

TABLE IX
MEASURED DC BIAS VALUES

Parameter	Value	Unit
V_{CC}	9.97	V
V_B	1.426	V
V_E	0.811	V
V_C	4.60	V
I_E	40.409	mA

B. Power and Efficiency

The DC input power is obtained from Eq. (13), which is Eq. 3.5 in the assignment. Since only the emitter current I_E was determined, it is used here as an approximation for I_C .

$$P_i \text{ (DC)} = V_{CC} \cdot I_C \quad (13)$$

The maximum theoretical output swing depends on the supply voltage V_{CC} and the DC bias point at the collector V_C and is given by Eq. (14). The expression gives V_o as voltage peak-to-peak.

$$V_o = \begin{cases} 2 \cdot V_C, & \text{if } V_C < \frac{1}{2} \cdot V_{CC} \\ 2 \cdot (V_{CC} - V_C), & \text{if } V_C > \frac{1}{2} \cdot V_{CC} \end{cases} \quad (14)$$

As V_C is less than half the supply voltage, the upper equation is used. Due to some small losses in the circuit the found voltage is also rounded down to whole volts.

The maximum AC output power is then estimated using Eq. (15), which is the peak-to-peak form of Eq. 3.6 in the assignment. This form is used since the calculated V_o represents the maximum peak-to-peak output voltage swing.

$$P_o \text{ (AC)} = \frac{V_o^2 \text{ (p-p)}}{8 \cdot R_C} \quad (15)$$

The efficiency is then calculated using Eq. (16), which is Eq. 3.7 in the assignment.

$$\eta = \frac{P_o}{P_i} \cdot 100 \quad (16)$$

The calculated values are listed in Table X.

TABLE X
CALCULATED POWER AND EFFICIENCY

Parameter	Value	Unit
V_o	9.0	V
P_i	406.23	mW
P_o	75.548	mW
η	18.60	%

For the experimental measurements, the input was increased until the maximum undistorted output was achieved, corresponding to the 40 mV rms setting of the signal generator. The resulting input and output voltages were measured, both as rms values as well. The collector resistor voltage drop is calculated as in Eq. (17), and the collector current from Ohm's law as in Eq. (18).

$$V_{RC} = V_{CC} - V_C \quad (17)$$

$$I_{RC} = \frac{V_{RC}}{R_C} \quad (18)$$

The measured input power is again calculated from Eq. (13), and the output power from Eq. (19), which this time uses the rms form of Eq. 3.6.

$$P_o \text{ (AC)} = \frac{V_o^2}{R_C} \quad (19)$$

The efficiency is then obtained from Eq. (16). All measured values are summarised in Table XI.

TABLE XI
MEASURED POWER AND EFFICIENCY

Parameter	Value	Unit
V_i	23.0	mV
V_o	2.38	V
P_i	399.48	mW
P_o	42.27	mW
η	10.58	%

Finally, the percentage differences between calculated and measured values of P_i , P_o , and η are given in Table XII, calculated using Eq. (6).

TABLE XII
RELATIVE DIFFERENCES BETWEEN CALCULATED AND MEASURED POWER AND EFFICIENCY

Parameter	Difference %	Unit
P_i	-1.66	%
P_o	-44.06	%
η	-43.11	%

IV. CONCLUSION

The investigation of BJT common-emitter and class-A power amplifiers showed that theoretical analysis can predict bias voltages, gain, and impedance with good accuracy. Most calculated values matched measurements within about ten per cent, confirming the reliability of the basic circuit equations. For the class-A amplifier, the measured efficiency was significantly lower than the calculated maximum, mainly due to non-ideal device behaviour and practical measurement limits. These results highlight the difference between ideal calculations and real performance, and show the importance of experimental verification in amplifier design.

ACKNOWLEDGEMENT

This report corresponds to Lab 03 in the course *Elektroniske enheter og kretser*. All related materials (L^AT_EX sources, figures, and code) are released under the MIT License and available at https://github.com/Kjelseth/EEK_lab.